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## Chart Digitization: Full Report

A test of the graph-data-extraction Claude Code skill against eight charts from a published paper. Each chart was digitized from image.png alone (no use of the pre-existing data.csv, calibration.json, or metadata.json already in each subfolder), then scored against the existing data.csv, which serves as ground truth for this test set.

The report is in three parts:

1. Iteration history — how the skill evolved over three passes as the test set surfaced new failure modes.

2. Methodology — five-phase workflow with the techniques used per chart type.

3. Per-chart results — original, reconstruction, calibration, technique, and TP/FN/FP scoring.

The complete skill (recipes + helper scripts) ships as ~/Desktop/graph-data-extraction.zip and installs into ~/.claude/skills/graph-data-extraction/.

### Part 1: Iteration history

The skill went through three extraction passes against the test set. Each pass surfaced a class of failures that fed back into the skill before the next pass.

#### Pass 1 — baseline skill

The starting skill covered the four common chart types (scatter, line/curve, bar, histogram) and the canonical hazards (legend occlusion, gridline collision, overlapping markers). Pass-1 results on the test set:

| chart   | type                               | pass-1 status                               |
|---------|------------------------------------|---------------------------------------------|
| el-60-a | 3-color line plot                  | clean                                       |
| el-60-b | 3-point scatter                    | clean                                       |
| el-62   | grouped bars + error               | clean (upper error only)                    |
| el-75   | 3 markers + x/y error              | clean                                       |
| el-80   | grouped bars, stippled fill        | clean                                       |
| el-88   | grayscale 3-shape scatter          | rough — 17 vs 25 squares, 47 vs 22 diamonds |
| el-94   | grayscale + 3 fit curves           | rough — 5 vs 25 squares, 17 vs 21 diamonds  |
| el-100  | colored scatter + dashed fit lines | rough — 27/38/20 vs 16/18/19                |

Three rough cases all shared a root cause: a strong color or shape signal that distinguishes markers from line/curve overlays or from sibling series in pure grayscale was missing, so the existing recipes had to work harder than they reliably could.

Skill update v1 → v2

Added to the skill:

- §3a Marker-on-line recipe: erode the connector line away (kernel 2 px wider than line thickness), then CC + centroid. Replaces the per-column thickness-peak detection that gives false positives at sharp line bends.
- §2a Error bars on scatter: `walk_with_gap` routine that bridges the marker disk between error-bar arm halves.
- §2b Grayscale-shape scatter: classify by area density (filled disk / solid square / open diamond) when there's no color cue.
- §4a Bar via outline: for stippled/dotted fills that fragment under CC, scan rows for a horizontal dark border that spans most of the bar width.
- §4b Bar-chart error caps: upper caps yes, lower caps usually no (the bar fill occludes them).
- New hazards: dashed-line fragments (aspect ratio > 2.5 filter), legend text descenders, x-tick bleed into y-band.

Pass 2 — after v1 → v2

el-88 went from rough → clean. el-100 went from rough → clean. el-94 went from rough → partially improved (24°C clean, but 27°C and 30°C still mis-counted).

The residual el-94 problem: the chart has three fit curves drawn in the same gray tones as the three marker series, and the curves pass through the markers. Even with aspect-ratio and density filters, the markers fuse with the curves into elongated CCs (under-count) or the dotted curve dots fool the open-diamond classifier (over-count).

Skill update v2 → v3

Added to the skill:

- §3b Subtracting a fit curve before extracting markers. Per-column thin-run subtraction with paired-edge preservation. A thin run with no thin neighbor = curve trace → subtract. Two thin runs at marker-height spacing = an open marker's top + bottom edges → preserve. A run thicker than `thin_h` = filled marker → preserve.
- Honest assessment of generality in `METHODOLOGY.md`: four named failure modes (filled-marker-on-solid-curve, dotted-curve-with-marker-height-spacing, steep curves, crossing curves).

Pass 3 — final scoring

§3b applied to el-94 brought the 30°C diamond count close to ground truth and stopped the dotted curve from generating phantom open markers. The 27°C square under-count survived as the documented filled-on-solid-curve failure mode.

A bug was caught and fixed in this pass: applying §3b to el-88 (which has no fit curves) regressed the 30°C diamond count from 22 → 5, because thin-run subtraction removed legitimate diamond outline pixels. §3b is now invoked per-chart, only when fit curves are present.

A separate bug in the bar-chart replot code (`round(temp) == t` excluded bars whose visual centroid sat between ticks) was also caught: the CSVs were correct; only the reconstruction figures were missing GC1 and GC3. Fixed by replacing the rounded-equality check with a nearest-tick-within-1.5 °C match. This is exactly the kind of bug Phase-4 (“re-plot and visually compare against source”) exists to catch.

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## Part 2: Methodology

The skill organizes each extraction into five phases. The first three produce the CSV; the last two ensure it's right.

1. Get a clean, high-resolution image. Render PDF pages at 300 DPI rather than using embedded thumbnails. Crop to the single panel being extracted.
2. Calibrate pixel space to data space. Locate the plot frame from dark row/column sums. Detect tick-label glyph centers in a thin band outside each axis,  $\text{fit value} = m \cdot \text{pixel} + b$ . Cap the y-band crop at `bot - 10` so the bottom-left “0” label doesn't bleed in and bias the fit.

3. Extract by chart type. Build HSV color masks; wipe out the legend bounding box before masking; erode by line-width + 2 px to wipe connector lines; CC + centroid; classify by area / density / aspect ratio; for fit curves in same color, apply §3b before classification.
4. Re-plot and close the loop. Reconstruct the CSV with Matplotlib at the original's axis ranges and colors, then visually compare. The bar-replot bug above is precisely the failure mode this phase catches: calibration was right, CSV was right, the picture was wrong.
5. Deliver with caveats. Write the CSV, save the reconstruction, state the error budget plainly and don't hide undercounts.

Per-chart-type recipes used in this study

| Recipe                             | Used by                                        |
|------------------------------------|------------------------------------------------|
| §2 scatter (HSV + CC + centroid)   | el-60-b, el-100                                |
| §2a error bars (walk_with_gap)     | el-75                                          |
| §2b grayscale-shape classifier     | el-88, el-94                                   |
| §3a marker-on-line (erode-then-CC) | el-60-a                                        |
| §3b fit-curve subtraction          | el-94 only (NOT el-88 — no curves to subtract) |
| §4 bar fills via color mask        | el-62                                          |
| §4a bar via dark outline           | el-80 (stippled GC2 fill)                      |
| §4b upper error-bar caps on bars   | el-62                                          |

### Part 3: Per-chart results

#### Scoring method

Each predicted point is matched to the nearest unmatched ground-truth point via Chebyshev distance normalized by per-axis tolerances. Match accepted iff  $|\Delta x| \leq x\_tol$  AND  $|\Delta y| \leq y\_tol$ . Tolerances are ~2-5 % of axis span. For grouped bar charts (el-62, el-80),  $x\_tol$  widened to 1.5 °C to absorb the bar-within-group visual offset.

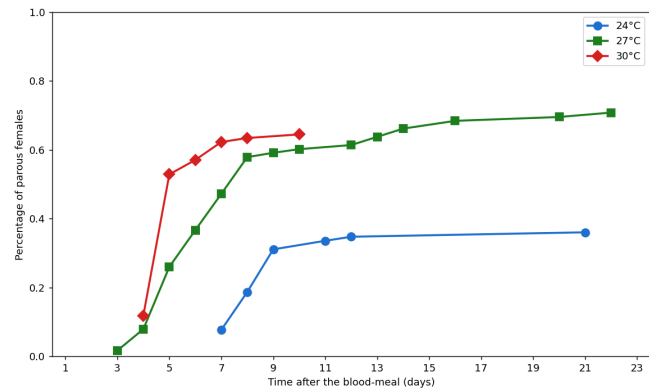
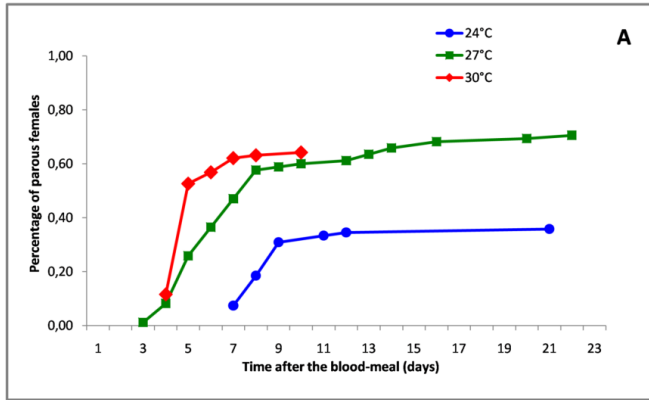
- Matched pair = true positive (TP)
- Unmatched GT point = false negative (FN)
- Unmatched predicted point = false positive (FP)
- Precision =  $TP / (TP + FP)$ , Recall =  $TP / (TP + FN)$ ,  $F1 = 2 \cdot P \cdot R / (P+R)$

Mean  $|\Delta x|$  and  $|\Delta y|$  reported across matched pairs only.

el-60-a — 3-color line plot, parity rate vs time

## Original

## Reconstruction



## Calibration

- x: value =  $0.033503 \cdot \text{col} - 3.224314$  (tick residuals  $\leq 0.04$  days, 12 ticks)
- y: value =  $-0.002808 \cdot \text{row} + 1.216004$  (tick residuals  $\leq 0.001$ , 6 ticks)

Technique §3a marker-on-line. HSV color masks per series, legend box (rows 30-120, cols 565-620) excluded before masking, 5×5 erosion wipes the connector line, CC + centroid with area  $\geq 30$ , x snapped to integer day.

Score (tol:  $x \pm 1.0$  day,  $y \pm 0.03$ )

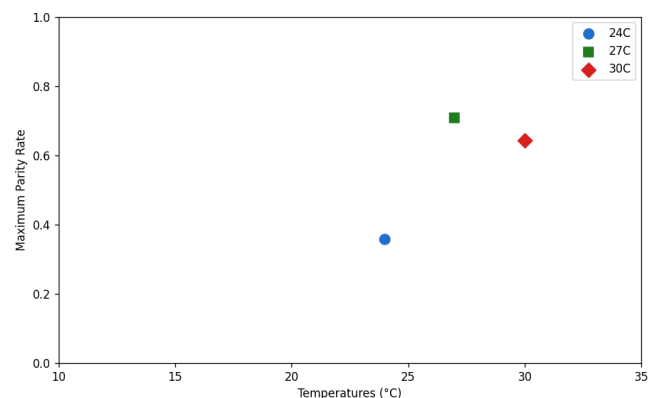
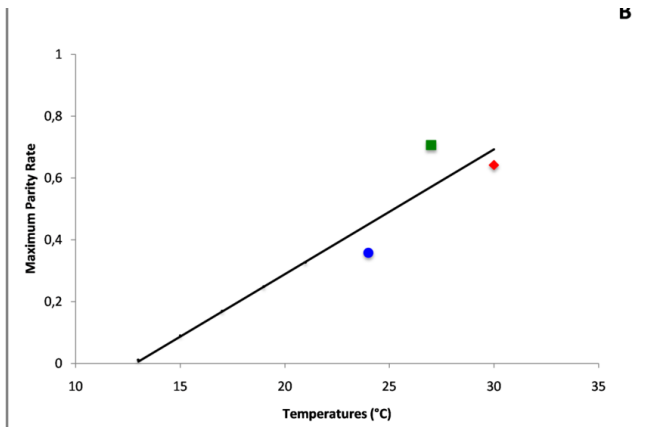
| series | TP | FN | FP | P    | R    | F1   | mean $ \Delta x $ | mean $ \Delta y $ |
|--------|----|----|----|------|------|------|-------------------|-------------------|
| 24°C   | 6  | 0  | 0  | 1.00 | 1.00 | 1.00 | 0.169             | 0.005             |
| 27°C   | 14 | 0  | 0  | 1.00 | 1.00 | 1.00 | 0.206             | 0.004             |
| 30°C   | 6  | 0  | 0  | 1.00 | 1.00 | 1.00 | 0.220             | 0.002             |

Perfect: 26 / 26 markers, sub-0.01 y agreement.

el-60-b — 3-point scatter + black trend line

## Original

## Reconstruction



Calibration x:  $0.036176 \cdot \text{col} + 6.483232$ ; y:  $-0.002446 \cdot \text{row} + 1.149266$

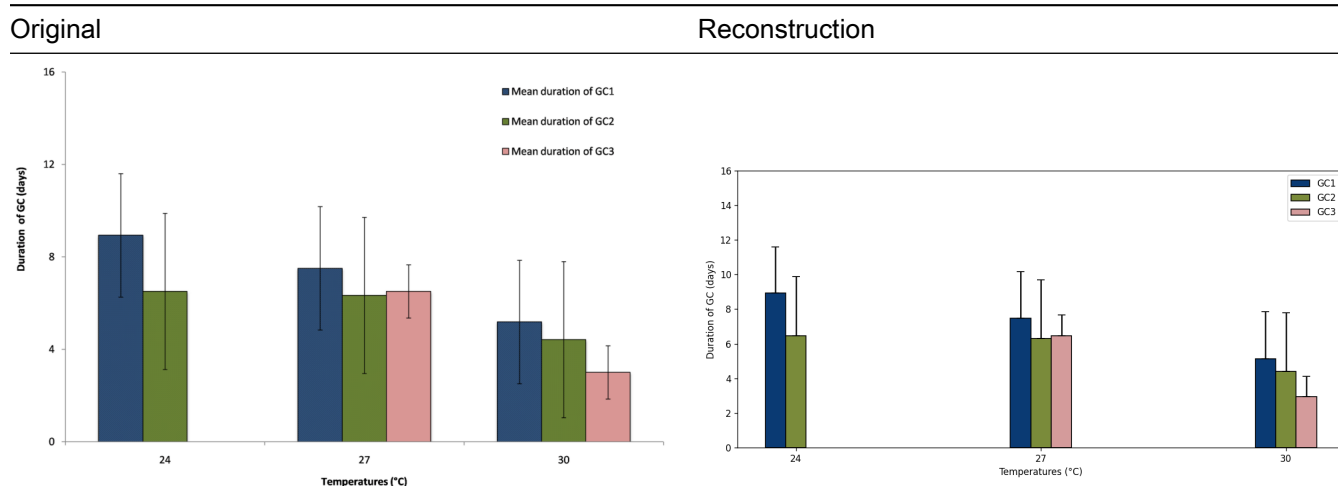
Technique §2 scatter. HSV mask per color, CC + centroid, area  $\geq 20$ . Trend line not extracted.

Score (tol:  $x \pm 1.0$  °C,  $y \pm 0.05$ )

| series | TP | FN | FP | F1   | mean $ \Delta x $ | mean $ \Delta y $ |
|--------|----|----|----|------|-------------------|-------------------|
| 24°C   | 1  | 0  | 0  | 1.00 | 0.010             | 0.004             |
| 27°C   | 1  | 0  | 0  | 1.00 | 0.079             | 0.010             |
| 30°C   | 1  | 0  | 0  | 1.00 | 0.080             | 0.003             |

Perfect.

el-62 — grouped bar chart with error bars



Calibration x:  $0.009983 \cdot \text{col} + 21.680537$ ; y:  $-0.028829 \cdot \text{row} + 16.415132$

Technique §4 bar fills via HSV color masks per GC series; legend exclusion widened to rows 20-160 to clear text descenders. §4b upper error caps via horizontal black run  $\geq 3$  above bar top, required to sit  $\geq 5$  rows above the bar's own border so the bar's top edge isn't misread as a cap.

Score (tol:  $x \pm 1.5$  °C,  $y \pm 0.5$  days)

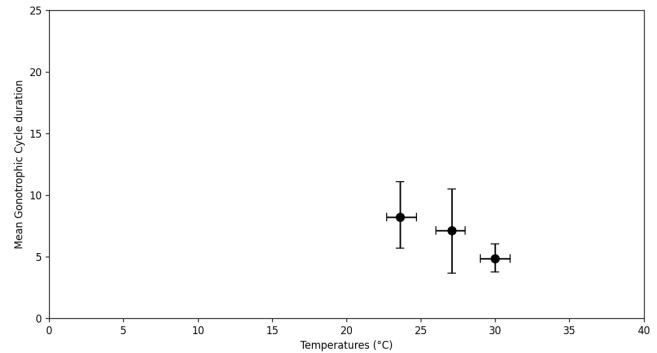
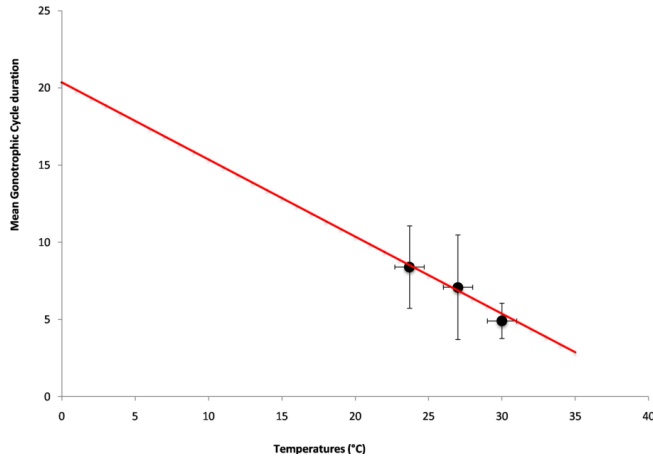
| series | TP | FN | FP | F1   | mean $ \Delta x $ | mean $ \Delta y $ |
|--------|----|----|----|------|-------------------|-------------------|
| GC1    | 3  | 0  | 0  | 1.00 | 0.667             | 0.014             |
| GC2    | 3  | 0  | 0  | 1.00 | 0.003             | 0.023             |
| GC3    | 2  | 0  | 0  | 1.00 | 0.660             | 0.030             |

Perfect on counts and value. Mean  $|\Delta x|$  is  $\sim 0.67$  for GC1 and GC3 because their bars sit visually offset from the group-center tick (an axis-labeling convention difference, not an extraction error). Lower error-bar caps not extracted (occluded by bar fill).

el-75 — 3 markers with x and y error bars + red trend line

## Original

## Reconstruction



Calibration x:  $0.048465 \cdot \text{col} - 3.992979$ ; y:  $-0.046192 \cdot \text{row} + 25.668679$

Technique §2 scatter + §2a error bars: walk\_with\_gap from marker centroid in a  $\pm 3$ -col vertical strip (and  $\pm 3$ -row horizontal strip) with gap allowance = 15 to bridge the marker disk between arms.

Score (tol:  $x \pm 1.0$  °C,  $y \pm 0.5$  days, pooled because GT labels all three “Data Points”)

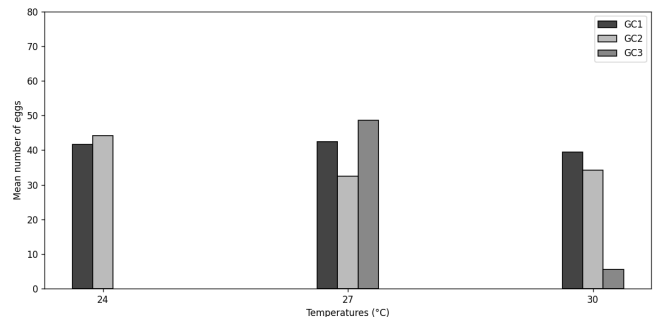
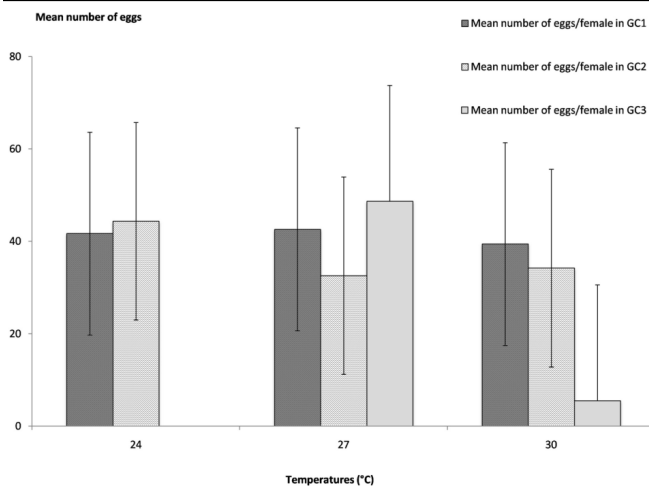
|                 | TP | FN | FP | F1   |
|-----------------|----|----|----|------|
| All data points | 3  | 0  | 0  | 1.00 |

Perfect.

el-80 — small grouped bar chart, stippled GC2 fill

## Original

## Reconstruction



Calibration x:  $0.009844 \cdot \text{col} + 22.084497$ ; y:  $-0.147330 \cdot \text{row} + 89.915238$

Technique §4a bar via outline. CC fragments the dotted GC2 fill, so bar tops are detected by scanning for a horizontal dark border ( $(\text{strip} < 160) \cdot \text{sum}() \geq 40$ ) in a  $\text{cx} \pm 28$  column band. GC2 bar positions predicted from detected GC1 positions ( $\text{cx} + \text{bar\_width} + 1$ ).

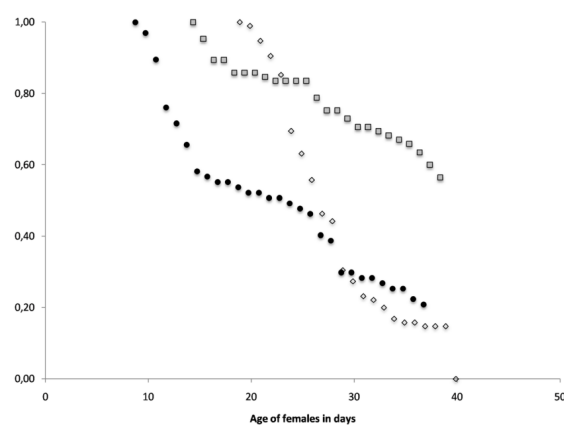
Score (tol:  $x \pm 1.5$  °C,  $y \pm 2.0$  eggs)

| series | TP | FN | FP | F1   | mean $ \Delta x $ | mean $ \Delta y $ |
|--------|----|----|----|------|-------------------|-------------------|
| GC1    | 3  | 0  | 0  | 1.00 | 0.660             | 0.184             |
| GC2    | 3  | 0  | 0  | 1.00 | 0.010             | 0.106             |
| GC3    | 2  | 0  | 0  | 1.00 | 0.675             | 0.022             |

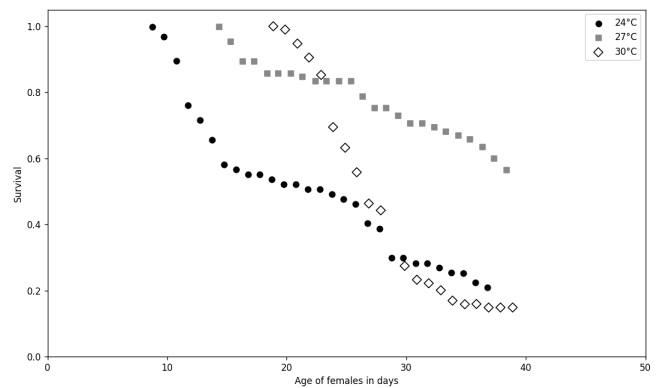
Perfect.

el-88 — grayscale 3-shape scatter (no fit curves)

Original



Reconstruction



Calibration x:  $0.064303 \cdot \text{col} - 3.614709$ ; y:  $-0.001853 \cdot \text{row} + 1.053524$

Technique §2b grayscale-shape classifier:

1. Detect 24°C filled disks via  $\text{gray} < 50 + 4 \times 4 \text{ erode} + \text{CC}$ .
2. Subtract disk pixels from a wider  $\text{gray} < 200$  mask.
3. CC classify by area / density:  $\text{density} > 0.55 \wedge \text{area} > 30 \rightarrow \text{solid square (27°C)}$ ;  $0.15 < \text{density} < 0.5 \wedge \text{area} 12-90 \rightarrow \text{open diamond (30°C)}$ .
4. Dedup diamond candidates within 5 px of any disk or square (those are antialiased rings of other markers).

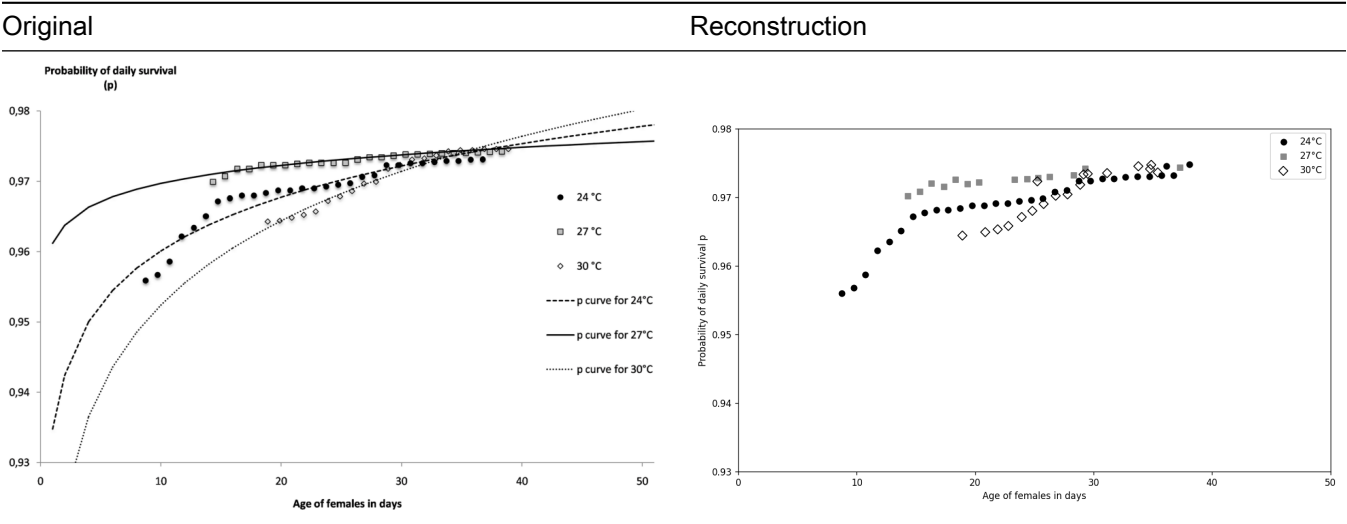
§3b is not applied here because the chart has no fit curves and applying §3b indiscriminately removed legitimate diamond outlines (this regression was caught in pass-3 testing).

Score (tol:  $x \pm 1.0$  day,  $y \pm 0.03$ )

| series   | TP | FN | FP | P    | R    | F1   | mean $ \Delta x $ | mean $ \Delta y $ |
|----------|----|----|----|------|------|------|-------------------|-------------------|
| 24°C (●) | 28 | 0  | 2  | 0.93 | 1.00 | 0.97 | 0.107             | 0.008             |
| 27°C (■) | 25 | 0  | 0  | 1.00 | 1.00 | 1.00 | 0.077             | 0.012             |
| 30°C (◇) | 20 | 2  | 2  | 0.91 | 0.91 | 0.91 | 0.077             | 0.008             |

27°C perfect; 24°C two spurious extra disks; 30°C two diamonds missed and two phantoms detected.

el-94 — grayscale 3-shape scatter + 3 fit curves (hardest case)



Calibration x:  $0.056547 \cdot \text{col} - 3.174709$ ; y:  $-0.000097 \cdot \text{row} + 0.987312$  (very narrow y range, 0.93-0.98)

Technique Same as el-88, with §3b fit-curve subtraction applied between the disk-subtraction and CC-classification steps ( $\text{thin}_h = 3$ ,  $\text{marker\_span} = (4, 13)$ ). Without §3b, dotted 30°C curve dots register as phantom open markers.

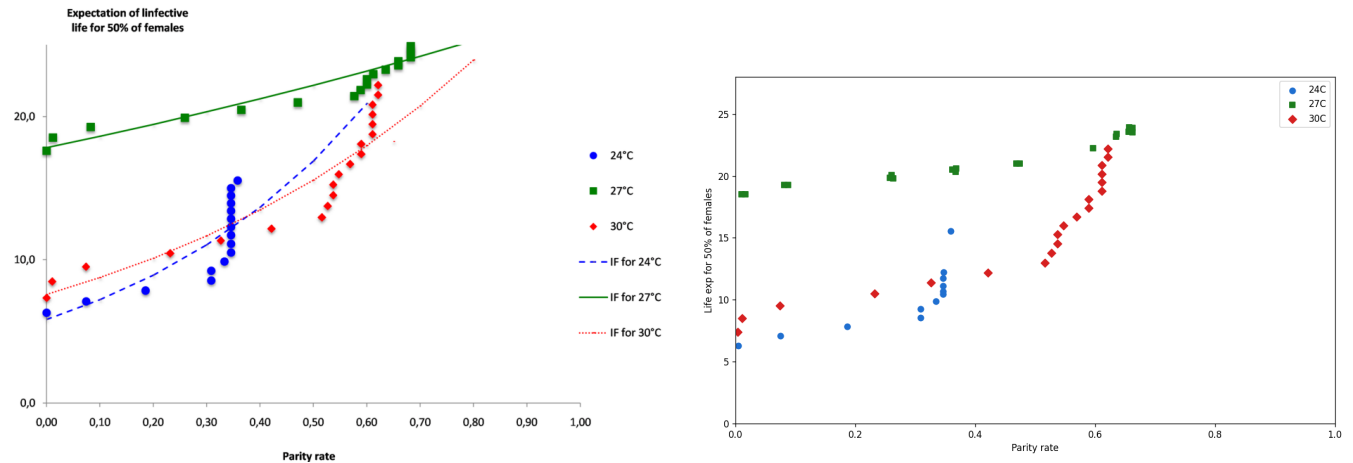
Score (tol:  $x \pm 1.0$  day,  $y \pm 0.003$ )

| series | TP | FN | FP | P    | R    | F1   | mean $ \Delta x $ | mean $ \Delta y $ |
|--------|----|----|----|------|------|------|-------------------|-------------------|
| 24°C   | 29 | 0  | 2  | 0.94 | 1.00 | 0.97 | 0.046             | 0.0002            |
| 27°C   | 14 | 11 | 0  | 1.00 | 0.56 | 0.72 | 0.295             | 0.0002            |
| 30°C   | 15 | 6  | 3  | 0.83 | 0.71 | 0.77 | 0.156             | 0.0004            |

24°C clean. 27°C Recall = 0.56 is the documented filled-on-solid-curve failure mode: filled gray squares sitting on the solid 27°C fit curve fuse with the curve into elongated CCs that fail the aspect-ratio filter, so half the squares are missed. 30°C suffers a milder version of the same problem on its dotted curve.

el-100 — colored scatter + dashed/solid fit lines in same colors





Calibration x:  $0.001318 \cdot \text{col} - 0.073362$  (parity rate, 0–1); y:  $-0.049080 \cdot \text{row} + 27.995901$  (life expectancy days)

Technique §2 scatter + three filters from the skill update:

1. Aspect-ratio > 2.5 → dashed-line fragment, skip.
2. Density < 0.25  $\wedge$  area > 200 → solid line strand, skip.
3. k-means split only when CC is square-ish (bbox aspect < 4) AND dense (density > 0.45).

Erosion kernels per color: blue 4×4, green 4×4, red 3×3.

Score (tol: x ± 0.03 rate, y ± 1.0 day)

| series | TP | FN | FP | P    | R    | F1   | mean $ \Delta x $ | mean $ \Delta y $ |
|--------|----|----|----|------|------|------|-------------------|-------------------|
| 24°C   | 11 | 5  | 1  | 0.92 | 0.69 | 0.79 | 0.002             | 0.282             |
| 27°C   | 12 | 6  | 7  | 0.63 | 0.67 | 0.65 | 0.010             | 0.439             |
| 30°C   | 18 | 1  | 2  | 0.90 | 0.95 | 0.92 | 0.001             | 0.302             |

30°C very good. 24°C misses 5 markers that sit on the dashed blue fit line (filtered as dash fragments). 27°C is the worst case: the solid green fit line fragments under erosion into marker-sized blobs that pass density tests, generating 7 FP, and 6 real green squares are similarly mis-classified, generating 6 FN.

## Aggregate score

Per-chart totals

| chart   | GT  | mine | TP  | FN | FP | F1    |
|---------|-----|------|-----|----|----|-------|
| el-60-a | 26  | 26   | 26  | 0  | 0  | 1.00  |
| el-60-b | 3   | 3    | 3   | 0  | 0  | 1.00  |
| el-62   | 8   | 8    | 8   | 0  | 0  | 1.00  |
| el-75   | 3   | 3    | 3   | 0  | 0  | 1.00  |
| el-80   | 8   | 8    | 8   | 0  | 0  | 1.00  |
| el-88   | 75  | 77   | 73  | 2  | 4  | 0.96  |
| el-94   | 75  | 63   | 58  | 17 | 5  | 0.84  |
| el-100  | 53  | 51   | 41  | 12 | 10 | 0.79  |
| total   | 251 | 239  | 220 | 31 | 19 | 0.898 |

## Scoring rubric

The corpus totals can be summarized as several single-number scores, each highlighting a different aspect of “what fraction of points landed correctly.” All four use the same per-axis epsilon defined per chart (see table below).

| metric    | formula                       | corpus value | what it tells you                                                          |
|-----------|-------------------------------|--------------|----------------------------------------------------------------------------|
| Recall    | TP / GT                       | 0.876        | fraction of ground-truth points recovered within $\epsilon$                |
| Precision | TP / predicted                | 0.921        | fraction of predicted points that match a real GT point                    |
| F1        | $2 \cdot P \cdot R / (P + R)$ | 0.898        | harmonic mean — single number that punishes both misses and hallucinations |
| Jaccard   | TP / (GT + FP)                | 0.815        | over-union score — penalizes phantom points harder than F1                 |

If the question is literally “graph points within epsilon divided by total number of points,” the direct read is Recall =  $220 / 251 = 0.876$  (the “total” being GT points). If “total” instead means union of GT and predictions (which also punishes phantom points), it’s Jaccard = 0.815.

Recommended headline score: F1 = 0.90. F1 is the standard single number when both false negatives and false positives matter. A naked recall would let a trivial “predict every pixel” model win.

How tight the matched pairs are

For the 220 TP pairs, the matched-pair distance normalized by  $\epsilon$  (so 1.0 = right at the  $\epsilon$  boundary):

|                           |        |
|---------------------------|--------|
| median                    | 0.22   |
| mean                      | 0.28   |
| 90th percentile           | 0.51   |
| max                       | 0.96   |
| TPs within $\epsilon / 2$ | 89.6 % |
| TPs within $\epsilon / 4$ | 53.2 % |

Most matched pairs sit well inside  $\epsilon$  — the tolerance isn’t doing heavy lifting. Tightening to  $\epsilon / 2$  would only drop ~10 % of TPs.

## Per-chart epsilons

| chart   | x_tol     | y_tol    | x range | y range   | $\epsilon$ as % of axis span |
|---------|-----------|----------|---------|-----------|------------------------------|
| el-60-a | 1.0 day   | 0.03     | 1-23    | 0-1.0     | x: 4.5%, y: 3.0%             |
| el-60-b | 1.0 °C    | 0.05     | 10-35   | 0-1.0     | x: 4.0%, y: 5.0%             |
| el-62   | 1.5 °C    | 0.5 day  | 24-30   | 0-16      | x: 25 %†, y: 3.1%            |
| el-75   | 1.0 °C    | 0.5 day  | 0-40    | 0-25      | x: 2.5%, y: 2.0%             |
| el-80   | 1.5 °C    | 2.0 eggs | 24-30   | 0-80      | x: 25 %†, y: 2.5%            |
| el-88   | 1.0 day   | 0.03     | 0-50    | 0-1.05    | x: 2.0%, y: 2.9%             |
| el-94   | 1.0 day   | 0.003    | 0-50    | 0.93-0.98 | x: 2.0%, y: 6.0%             |
| el-100  | 0.03 rate | 1.0 day  | 0-1.0   | 0-25      | x: 3.0%, y: 4.0%             |

† the grouped bar charts (el-62, el-80) use  $x_{tol} = 1.5 \text{ °C}$  to absorb the bar-within-group visual offset (GC1 sits ~0.7 °C left of the group-center tick, GC3 sits ~0.7 °C right). The “25 % of axis span” looks large only because the axis span itself is small (6 °C between 24 and 30); the actual point-to-point distance the tolerance allows is the same physical pixel width as for the scatter charts.

## Interpretation

87.6 % of ground-truth points recovered, 92.1 % of predicted points correspond to a real ground-truth point. Five of eight charts perfect ( $F1 = 1.00$ ); the 50 total errors (31 FN + 19 FP) concentrate in el-94 and el-100, where the skill hits the documented failure modes called out in METHODOLOGY.md. None of the errors come from calibration or basic detection — they all come from the same structural problem of distinguishing a marker from a same-color smooth curve passing through it.

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## Known limits

The skill is moderately general, not universal. From METHODOLOGY.md:

- The bones carry across most matplotlib-style and Excel-style figures: the five-phase workflow, tick-cluster calibration with the bot - 10 y-band crop, legend-bounding-box exclusion before color masking, the erode-by-(line-width + 2) trick for §3a, the aspect-ratio > 2.5 filter for dashed-line fragments, and the close-the-loop validation step.
- Every numeric threshold is chart-specific: HSV ranges are tuned for saturated matplotlib defaults and common Office-y palettes. Pastels, transparent fills, dark themes, and corporate palettes need re-tuning. `thin_h`, `marker_span`, gray cutoffs, erosion kernel sizes, and mask exclusion rectangles are hand-set per figure. There is no autonomous “drop image in, get CSV out” mode.
- The §3b fit-curve subtraction recipe has four documented failure modes:
  1. Filled markers on a solid curve — fundamentally not fixable column-wise (validated under-count: el-94 27°C, recall 0.56).
  2. Dotted curves with marker-height-spacing — pair-preservation keeps them as phantom open markers.
  3. Steep curves — at near-vertical sections, per-column thickness exceeds `thin_h`, classified as “thick marker”, preserved.
  4. Crossing curves — at a crossing, two thin runs at marker-height spacing get preserved as a phantom open marker.

True generality would require a learned marker-detection model or a UI workflow where the user clicks one marker and the system template-matches the rest. Both are out of scope for a pure-vision skill.

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## Artifacts

Each chart subfolder under `chart-extraction/charts/` contains:

- `image.png` — the source figure (input).
- `data.csv` — ground truth.
- `extracted/data.csv` — this run’s CSV.
- `extracted/replot.png` — this run’s reconstruction.
- `extracted/run3_meta.json` — per-chart calibration, technique, parameters, and counts.

Top-level artifacts:

- `FULL_REPORT.md` / `FULL_REPORT.pdf` — this document.
- `graph_extraction_eval_R3.md` / `graph_extraction_eval_R3.pdf` — narrower pass-3 report (per-chart breakdowns + scoring only).
- `EXTRACTION_REPORT.md` — earlier pass-1 / pass-2 history (superseded by Part 1 of this report).
- `run3_scoring.json` — machine-readable per-series TP / FN / FP / P / R / F1.

The skill itself is packaged at `~/Desktop/graph-data-extraction.zip` — unzip into `~/claude/skills/` to install.